Recursive CTEs, in five diagrams

A recursive CTE is not Java-style recursion. It is a **while loop**: the anchor runs once, the recursive part runs on only the previous round's rows, and the loop stops when a round produces no rows. The outer query then runs once on everything collected.

lastBatch · the CTE name inside the recursive part

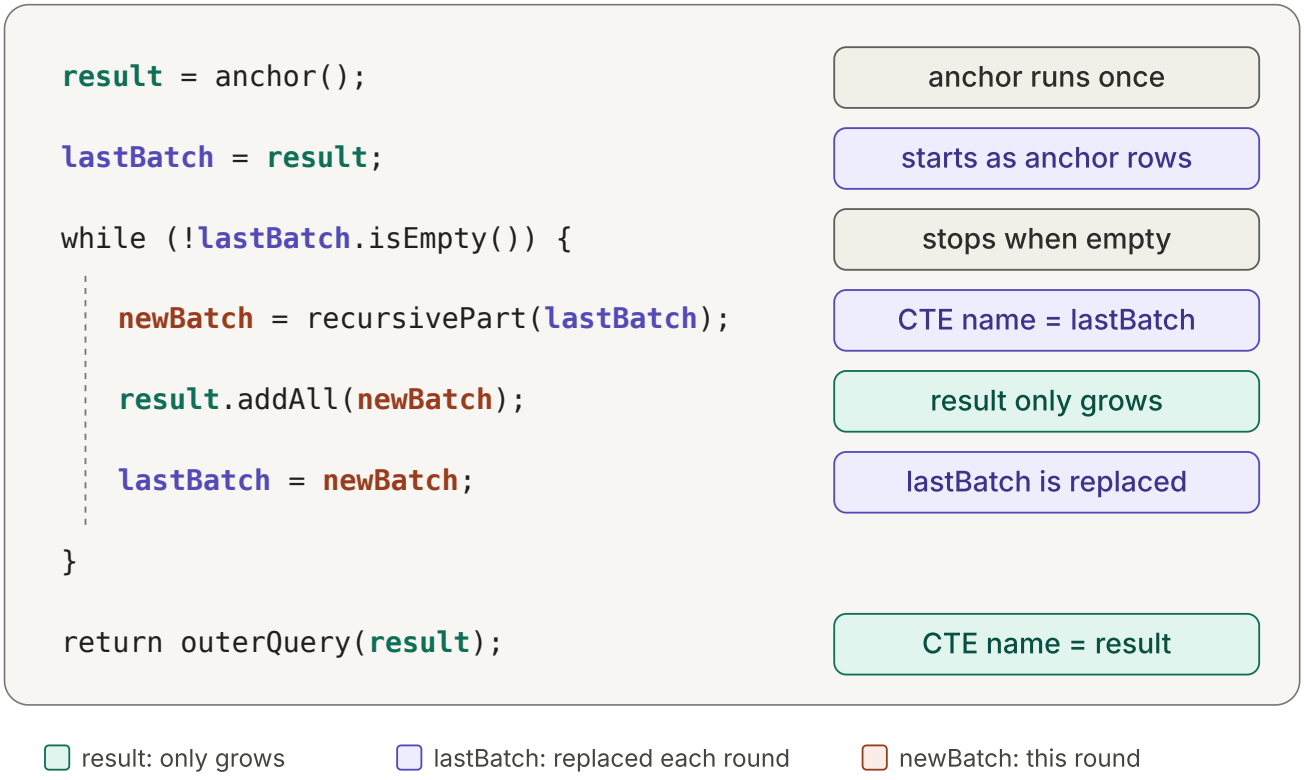
newBatch · the rows made this round

result · everything collected so far

Gray · code blocks: anchor, recursivePart, outerQuery

1 · The loop cheat sheet

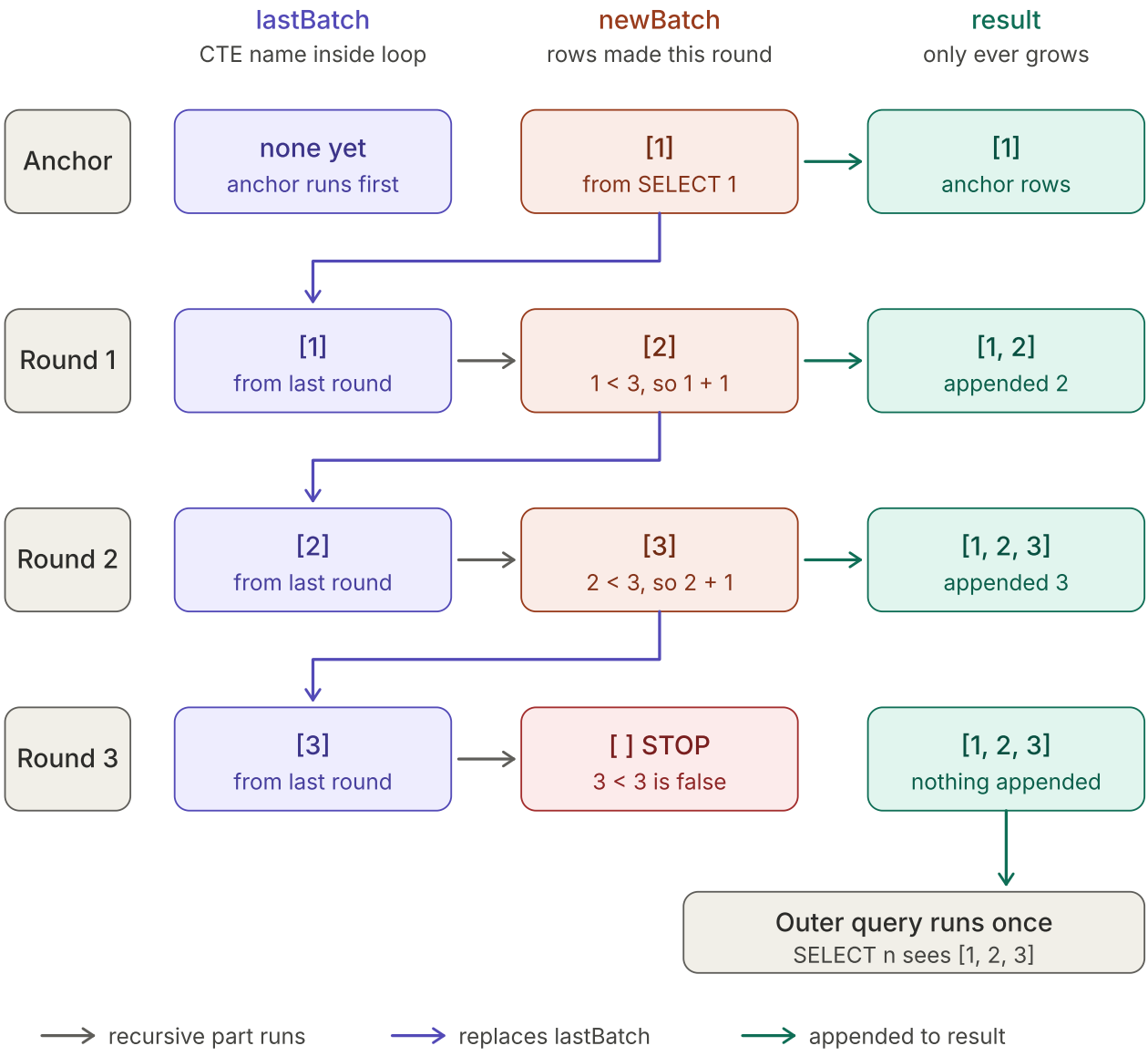
What the CTE name means on each line of the loop.



1. The loop cheat sheet (this page)
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2 · Counting dry run, 1 to 3

One row per round. The anchor makes `[1]`, then each round adds one number until the filter `n < 3` fails.

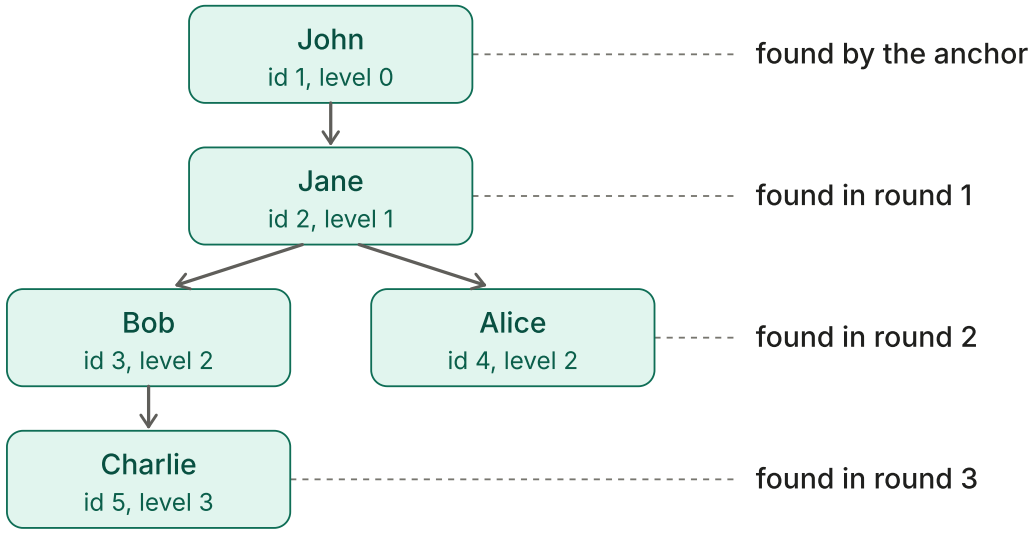


Read each row left to right: `lastBatch` goes into the recursive part, which makes `newBatch`, which is appended to `result`. The purple arrow shows `newBatch` becoming the next round's `lastBatch`. The red box is the stop: that round produced nothing, so the loop ends.

3 · Employee data: table and tree

Each row stores its manager's id in `manager_id`. The tree shows the same data, with the round in which the loop finds each person.

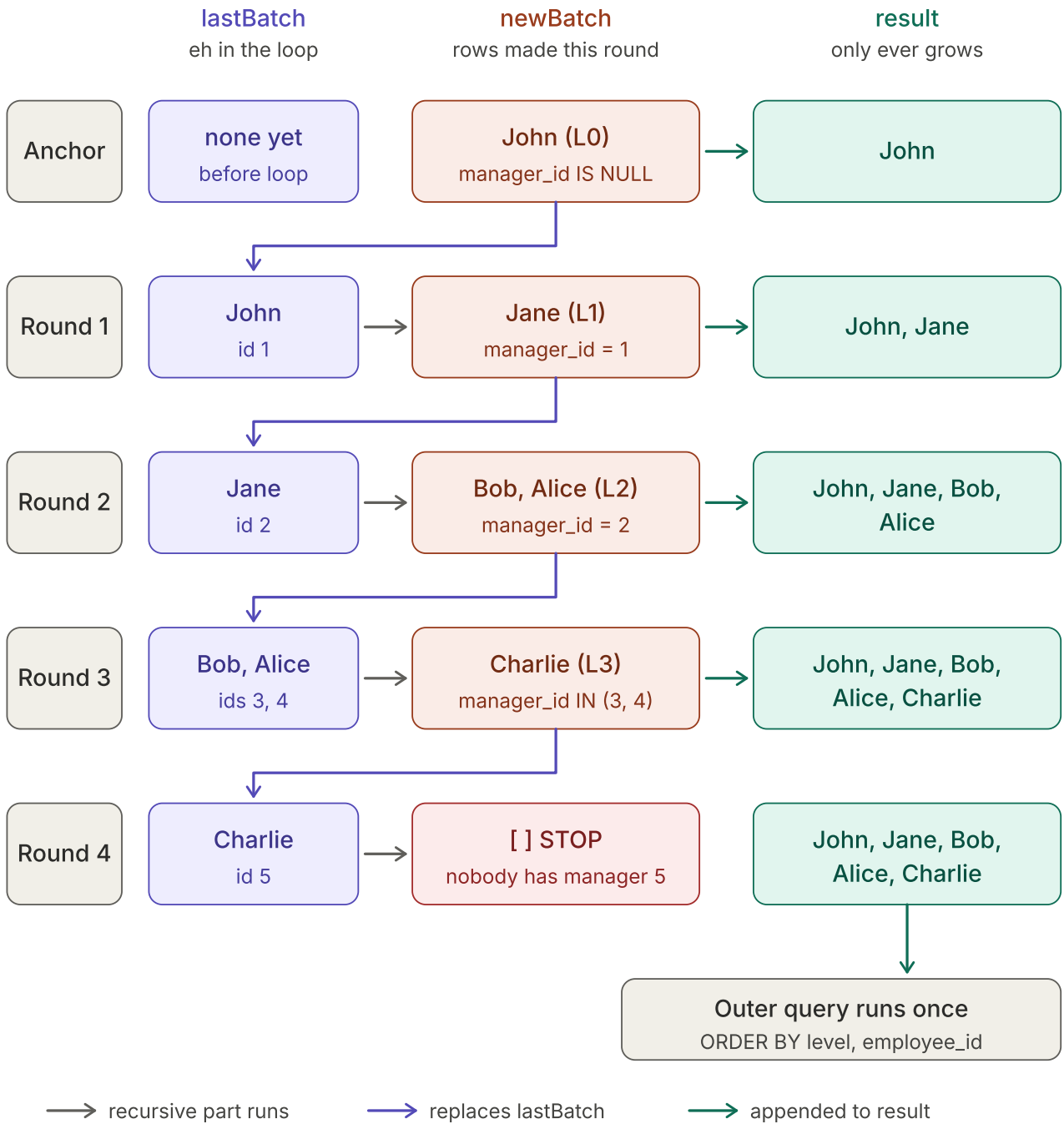
employee_id	employee_name	manager_id
1	John Doe	NULL
2	Jane Smith	1
3	Bob Johnson	2
4	Alice Brown	2
5	Charlie Davis	3



Each level of the tree is one round of the loop. `level` is not stored in the table: the anchor sets it to 0 and each round adds 1 (`eh.level + 1`), so the round number equals the level.

4 · Employee hierarchy dry run

The recursive part asks one question each round: who has a `manager_id` equal to an id in `lastBatch`? Each round walks one level down the org chart.



Round 4 finds nobody with manager 5, so the loop stops. The outer `ORDER BY level, employee_id` runs once, on all five rows.

5 · Syntax map

The same keywords on the general syntax and on both example queries. Each tinted row links a code line to its label.

